

Forecasting Germany's Consumer Price Index (CPI) using Time Series Models

BA26-04 Time Series | Project Proposal

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Introduction

Project Scope

This project focuses on forecasting Germany's inflation using the Consumer Price Index (CPI) as a time series.

- The main objective is to predict future CPI values from historical monthly observations.
- The analysis focuses on patterns such as trend, seasonality, and autocorrelation.
- Classical forecasting models, ARIMA, ETS, and SARIMA, are compared to assess how well they capture inflation dynamics.

Motivation

Motivation

- CPI measures the average change in prices of a typical basket of goods and services over time.
- CPI is one of the main indicators used to track inflation in Germany.
- Forecasting CPI helps us understand how inflation may develop in the near future.
- This is useful for economic analysis, planning, and comparing changes over time.
- The project also shows how time series models can turn past inflation data into interpretable forecasts.

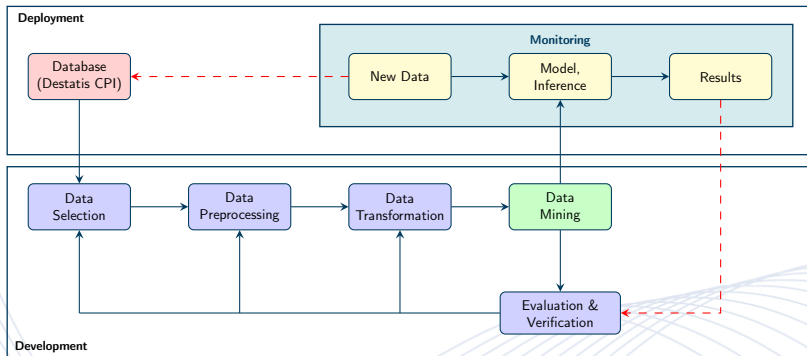
For more on the basket of goods, see: <https://www.destatis.de/EN/Themes/Economy/Prices/Consumer-Price-Index/BasketGoodsWeightingPattern.html>

Objectives

- Forecast future CPI values from historical monthly observations.
 - Prepare the CPI data so it is stable over time and suitable for forecasting models.
 - Identify and interpret long-term trend and possible seasonal effects.
- Compare and evaluate models and identify which best captures Germany's CPI dynamics.

Methodology

KDD-Based Workflow



- The lower part shows the core KDD development workflow from data selection to evaluation, while the upper part extends it into deployment through monitoring, new incoming data, model inference, results, and feedback loops for future updates.

Dataset

Dataset

- Source: Destatis GENESIS database, table 61111-0002.
- Link: <https://www-genesis.destatis.de/datenbank/online/statistic/61111/table/61111-0002>
- Monthly CPI observations for Germany from 1991 to 2026.
- Core fields: Date and CPI.
- With 2020 = 100, values below 100 mean lower prices than in 2020.
- Suitable for classical forecasting because the series is chronological and evenly spaced.

Dataset

Consumer price index: Germany, months

Available period: 1991 / 2026

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Consumer price index for Germany
Germany

Year Months	Consumer price index		Change on previous year's month		Change from previous month	
		2020=100		in (%)		in (%)
2025	January	120.3		2.3		-0.2
	February	120.8		2.3		0.4
	March	121.2		2.2		0.3
	April	121.7		2.1		0.4
	May	121.8		2.1		0.1
	June	121.8		2.0		-
	July	122.2		2.0		0.3
	August	122.3		2.2		0.1
	September	122.6		2.4		0.2
	October	123.0		2.3		0.3
	November	122.7		2.3		-0.2
	December	122.7		1.8		-
2026	January	122.8		2.1		0.1
	February	123.1		1.9		0.2

Figure: Preview of the Germany CPI dataset. Source:

<https://www-genesis.destatis.de/datenbank/online/statistic/61111/table/61111-0002>

Data Transformation

Data Transformation

- We break the CPI data into trend, seasonality, and random variation.
- We use the ADF test to check whether the data is stable over time.
- If the data is not stable, differencing is used by comparing each value with the previous one.
- ACF and PACF plots help us see how current values relate to past values.
- The goal is to prepare the data so forecasting models can work better.

Data Mining

Data Mining

- In this phase, we train forecasting models on the prepared CPI data.
- ARIMA uses past values and past errors to make forecasts.
- ETS models the level, trend, and seasonality of the series directly.
- SARIMA is similar to ARIMA, but also includes seasonal patterns.
- The goal is to compare which model forecasts future CPI values best.

Evaluation

Evaluation

Since the objective is forecasting continuous values, regression-based metrics are used.

- RMSE captures overall prediction error and penalizes large deviations.
- MAE provides a stable and interpretable average error.
- MAPE allows comparison in percentage terms, making results easier to interpret.

Metric	Full Form	Description
RMSE	Root Mean Squared Error	Penalizes larger errors more heavily
MAE	Mean Absolute Error	Average absolute difference between actual and predicted values
MAPE	Mean Absolute Percentage Error	Expresses error as a percentage

Expected Output

Expected Output

- A ranked comparison of ARIMA, ETS, and SARIMA on German CPI forecasting.
- Interpretable findings on trend, seasonality, and model suitability.
- A reproducible workflow that can be updated as new CPI data becomes available.

Model	RMSE	MAE	MAPE
ARIMA	TBD	TBD	TBD
ETS	TBD	TBD	TBD
SARIMA	TBD	TBD	TBD

Potential Delighters

Potential Delighters

- Add forecast confidence intervals to quantify uncertainty.
 - Compute prediction intervals, Eg. 80% and 95% for all models.
 - Visualize upper and lower bounds together with the forecasts.
 - Compare confidence intervals across ARIMA, ETS, and SARIMA.
- Investigate structural breaks or inflation shocks in the CPI series.
 - Identify periods of sudden change, such as COVID-19/energy crisis.
 - Analyze deviations from the long-term inflation trend.
 - Observe model performance before and after structural breaks.
 - Interpret possible economic reasons behind these shocks.

Keyword Full Forms

- CPI: Consumer Price Index
- ADF: Augmented Dickey-Fuller
- Stationarity: a stable time-series property
- ACF: Autocorrelation Function
- PACF: Partial Autocorrelation Function
- ETS: Error, Trend, Seasonality
- ARIMA: AutoRegressive Integrated Moving Average
- SARIMA: Seasonal AutoRegressive Integrated Moving Average

Keyword Definitions

- CPI: the inflation indicator being forecast in this project.
- ADF: statistical test used to check whether the time series is stationary.
- Stationarity: stable statistical behavior over time, required by ARIMA-type models.
- Decomposition: separation of the time series into trend, seasonal, and residual components.
- Differencing: transformation that subtracts previous values to remove trend and improve stationarity.
- ACF: diagnostic tool for temporal correlation across lags.
- PACF: diagnostic tool that helps identify direct lag relationships and model order.
- ETS, ARIMA, SARIMA: the three forecasting model families compared in the project.

Thank you!
Questions and discussion